

TANISH PAL

Software Engineer — Full-Stack Development — Analyst — C++ — Python — DSA

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EDUCATION

Dr. B. R. Ambedkar National Institute of Technology (NIT Jalandhar)

2022 – 2026

Bachelor of Technology in Electronics and Communication Engineering

EXPERIENCE

Basil Infotech

July – Dec 2025

Software Development Intern

Remote

- Engineered a scalable CRM and ERP system integrating client management, centralized data processing, and automated reporting to improve operational efficiency.
- Designed and implemented key modules including Client Database and Lead Management, enabling efficient client operations and streamlined workflows.

A2IT

June – July 2025

Software Development Intern

Remote

- Built 10+ responsive React (CRA) UI components for Cheffer, improving cross-device compatibility; optimized state management to reduce unnecessary re-renders and improve performance.
- Integrated REST APIs in collaboration with the backend team, ensuring seamless data flow between client and server.

PROJECTS

Mock Mate 🔗 | *Next.js 14, Gemini API, Neon PostgreSQL, Drizzle ORM, Web Speech API, TailwindCSS* **Live Demo** 🔗

- Engineered an automated AI technical assessment platform utilizing **Google Gemini LLM**, providing instantaneous multi-metric scoring, model answers, and qualitative feedback across customized technical domains.
- Integrated browser-native **Web Speech STT/TTS engines** to achieve sub-second conversational voice latency, enabling hands-free, interactive vocal interview simulations without third-party audio service costs.
- Streamlined data persistence and query latency via **Neon Serverless PostgreSQL** and **Drizzle ORM**, managing multi-stage interview state, structured question banks, and historical performance analytics.
- Hardened application security posture by decoupling client bundles from database credentials via **React Server Actions**, achieving **0% credential leakage** and robust role-based route authentication via **Clerk**.

Core Banking System 🔗 | *C++, STL, File I/O, Git, lld, OOPS*

Live Demo 🔗

- Built a PIN-authenticated core banking system in C++17 supporting account creation, deposits, withdrawals, and inter-account transfers with real-time balance validation.
- Designed a custom file-based persistence layer with an append-only audit log, ensuring transaction history and account state reliably survive application restarts.
- Engineered a cross-platform CLI with adaptive terminal rendering (color/UTF-8 detection) and automated launcher scripts, compiling warning-free under **-Wall -Wextra -Wpedantic**.

KnowYourProduct 🔗 | *React, Node.js, Express, MongoDB, Tailwind CSS*

Live Demo 🔗

- Engineered a full-stack health and nutrition platform using React and Node.js, enabling consumers to scan, search, and compare packaged foods for detailed dietary insights.
- Integrated a real-time browser-based barcode scanner using the HTML5-QRCode API, mapping physical products to detailed analyses of nutritional metrics, allergens, and ingredient concern levels.
- Architected a secure RESTful API with Express and MongoDB, leveraging Mongoose text indexing for rapid product search and JWT authentication for personalized user dashboards.

Sales Performance Dashboard 🔗 | *Power BI, Power Query, DAX, Excel*

Live Demo 🔗

- Built an interactive Power BI dashboard to analyze sales performance across products, regions, customers, and time periods using a structured sales dataset.
- Cleaned and transformed raw data using Power Query and created calculated measures in DAX to track key performance indicators including sales, profit, quantity, and growth.
- Designed interactive visualizations including KPI cards, bar charts, line charts, pie charts, slicers, and filters to enable dynamic exploration of business performance.

TECHNICAL SKILLS

Languages: C/C++, Python, JavaScript (ES6+), TypeScript, SQL

Frameworks & Libraries: React.js, Redux, Next.js, Node.js, Express.js, TailwindCSS

Python & Automation: Pytest, Requests, REST APIs, JSON, Logging, Test Automation, API Testing

Databases: MySQL, PostgreSQL, MongoDB

Tools & APIs: Git, GitHub, Postman, Vercel, VS Code, OpenAI API, Google GenAI API

Data Analysis & Visualization: Power BI, Power Query, DAX, Excel, Pandas, NumPy, Matplotlib, Tableau

Coursework: Data Structures & Algorithms, OOP, Operating Systems, DBMS